

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM} + S_1 + S_2$. The report combines the renormalisable RGBeta running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q}Y_u \tilde{H}u + \text{h.c.}$
Y_d	$-\bar{Q}Y_d H d + \text{h.c.}$
Y_e	$-\bar{L}Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$

$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$$\beta_{g_1}^{(1)}$$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$$

Column Representative term structure

T1	$\frac{55g_1^3}{6}$
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Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{55g_1^3}{6}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{43g_1^3}{6}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{47g_1^3}{6}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{17g_1^3}{2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{43g_1^3}{6}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{17g_1^3}{2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{17g_1^3}{2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{43g_1^3}{6}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{17g_1^3}{2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{47g_1^3}{6}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{43g_1^3}{6}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{55g_1^3}{6}$

Model	α	S_1	S_2	F	T1
T3-E-m4	-4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$\frac{71g_1^3}{6}$
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$\frac{47g_1^3}{6}$
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$\frac{47g_1^3}{6}$
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$\frac{71g_1^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure			
T1	$-\frac{5g_2^3}{2}$			

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{5g_2^3}{2}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{5g_2^3}{2}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{5g_2^3}{2}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-\frac{17g_2^3}{6}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-\frac{17g_2^3}{6}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-\frac{17g_2^3}{6}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-\frac{17g_2^3}{6}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-\frac{17g_2^3}{6}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-\frac{17g_2^3}{6}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-\frac{5g_2^3}{2}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-\frac{5g_2^3}{2}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-\frac{5g_2^3}{2}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{11g_2^3}{6}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{11g_2^3}{6}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{11g_2^3}{6}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-\frac{11g_2^3}{6}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\,\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure			
T1	$-7g_3^3$			

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-7g_3^3$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-7g_3^3$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-7g_3^3$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-7g_3^3$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-7g_3^3$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-7g_3^3$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-7g_3^3$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-7g_3^3$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-7g_3^3$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-7g_3^3$

$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column		Representative term	structure
T1	3	$\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	
T2	+	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	
T3	+3	$\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column		Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	
T3	$-8g_3^2\ (Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column		Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A-m2	-2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A-p0	0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(1, -1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(3, -1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-m2	-2	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-p0	0	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-p2	2	$(3, 1)$	$(1, 2)$	$(2, \frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-m4	-4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$
Quantity: Y_d

$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column		Representative term structure
T1		$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$
T2		$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$
T3		$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) (Y_d)_{ij}$	$+3\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) (Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column		Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	
T3	$-8g_3^2\ (Y_d)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column		Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column		Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	
T2	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	
T3	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger}Y_e\right)_{ij}$	$-\frac{9}{4}g_2^2\left(Y_e\right)_{ij}$	$-\frac{15}{4}g_1^2\left(Y_e\right)_{ij}$

$$\beta_{m_1^2}^{(1)}$$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$$

Terms 1 of 3

Column	Representative term structure	
T1	$-24g_1^2m_1^2$	
T2	$+6\lambda_{12}^{(1)}m_2^2$	
T3	$+4\lambda_{S_1}^{(1)}m_1^2$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-24g_1^2m_1^2$	$+6\lambda_{12}^{(1)}m_2^2$	$+4\lambda_{S_1}^{(1)}m_1^2$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-6g_1^2m_1^2$	$+6\lambda_{12}^{(1)}m_2^2$	$+4\lambda_{S_1}^{(1)}m_1^2$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	$6\lambda_{12}^{(1)}m_2^2$	$+4\lambda_{S_1}^{(1)}m_1^2$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	$+4\lambda_{12}^{(1)}m_2^2$	$+6\lambda_{S_1}^{(1)}m_1^2$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-6g_1^2m_1^2$	$+2\lambda_{12}^{(1)}m_2^2$	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	$+2\lambda_{12}^{(1)}m_2^2$	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-6g_1^2m_1^2$	$+2\lambda_{12}^{(1)}m_2^2$	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-24g_1^2m_1^2$	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-6g_1^2m_1^2$	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-6g_1^2m_1^2$	—	—

Quantity: m_1^2

Terms 2 of 3

Column		Representative term structure
T1	$-\frac{9g_2^2m_1^2}{2}$	
T2	$-\frac{27g_1^2m_1^2}{2}$	
T3	$-12g_2^2m_1^2$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{27g_1^2m_1^2}{2}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{27g_1^2m_1^2}{2}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$-12g_2^2m_1^2$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$-12g_2^2m_1^2$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$-12g_2^2m_1^2$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-12g_2^2m_1^2$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-12g_2^2m_1^2$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-12g_2^2m_1^2$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-12g_2^2m_1^2$

Quantity: m_1^2

Terms 3 of 3

Column	Representative term structure	
T1	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	
T2	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$$

Terms 1 of 3

Column	Representative term structure		
T1	$-12g_2^2m_2^2$		
T2	$-6g_1^2m_2^2$		
T3	$+2\lambda_{12}^{(1)}m_1^2$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	$+2\lambda_{12}^{(1)}m_1^2$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-12g_2^2m_2^2$	—	$+2\lambda_{12}^{(1)}m_1^2$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	$+2\lambda_{12}^{(1)}m_1^2$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$+4\lambda_{12}^{(1)}m_1^2$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$6\lambda_{12}^{(1)}m_1^2$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-6g_1^2m_2^2$	$+6\lambda_{12}^{(1)}m_1^2$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-24g_1^2m_2^2$	$+6\lambda_{12}^{(1)}m_1^2$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-12g_2^2m_2^2$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-12g_2^2m_2^2$	$-24g_1^2m_2^2$	—

Quantity: m_2^2

Terms 2 of 3

Column	Representative term structure	
T1	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	
T2	$-\frac{9g_2^2m_2^2}{2}$	
T3	$-\frac{3g_1^2m_2^2}{2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{27g_1^2m_2^2}{2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{27g_1^2m_2^2}{2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	—

Quantity: m_2^2

Terms 3 of 3

Column	Representative term structure	
T1	$+6\lambda_{S_2}^{(1)}m_2^2$	
T2	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	$+6\lambda_{S_2}^{(1)}m_2^2$	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	$+4\lambda_{S_2}^{(1)}m_2^2$	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	$+4\lambda_{S_2}^{(1)}m_2^2$	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	$+4\lambda_{S_2}^{(1)}m_2^2$	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	—	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	—	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	—	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Quantity: λ_1

Terms 1 of 7

[illegible]

Quantity: λ_1

Terms 2 of 7

Column	Representative term structure
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$
T2	$+4\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 7

Column		Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	
T2	$-9g_2^2\lambda_1$	
T3	$+\frac{3g_1^4}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Quantity: λ_1

Terms 4 of 7

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+2\lambda_{HS_1}^{(1)2}$
T3	$+6\lambda_{HS_2}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$

Quantity: λ_1

Terms 5 of 7

Column		Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$	
T2	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	
T3	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—

Quantity: λ_1

Terms 6 of 7

Column		Representative term structure
T1		$+8\lambda_7\left(\lambda_7\right)^*$
T2		$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3		$+8\lambda_8\left(\lambda_8\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+8\lambda_8\left(\lambda_8\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+8\lambda_8\left(\lambda_8\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_1

Terms 7 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 8

Column		Representative term structure
T1		$192g_1^4$
T2		$-48g_1^2\lambda_{S_1}^{(1)}$
T3		$+6\lambda_{12}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$192g_1^4$	$-48g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$12g_1^4$	$-12g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	$6\lambda_{12}^{(1)2}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	$+4\lambda_{12}^{(1)2}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$+12g_1^4$	—	$+2\lambda_{12}^{(1)2}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	$+2\lambda_{12}^{(1)2}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+12g_1^4$	—	$+2\lambda_{12}^{(1)2}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+192g_1^4$	—	$+6\lambda_{12}^{(1)2}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+12g_1^4$	—	$+6\lambda_{12}^{(1)2}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	$+6\lambda_{12}^{(1)2}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+12g_1^4$	—	$+6\lambda_{12}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 8

Column	Representative term structure	
T1	$+4\lambda_{HS_1}^{(1)2}$	
T2	$+10\lambda_{S_1}^{(1)2}$	
T3	$\frac{9g_2^4}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+10\lambda_{S_1}^{(1)2}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+10\lambda_{S_1}^{(1)2}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+10\lambda_{S_1}^{(1)2}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$\frac{9g_2^4}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$\frac{9g_2^4}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$\frac{9g_2^4}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{HS_1}^{(1)2}$	$+12\lambda_{S_1}^{(1)2}$	$\frac{9g_2^4}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_1}^{(1)2}$	$+14\lambda_{S_1}^{(1)2}$	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 8

Column		Representative term structure
T1	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$	
T2	$-9g_2^2\lambda_{S_1}^{(1)}$	
T3	$+\frac{243g_1^4}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{243g_1^4}{4}$
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{243g_1^4}{4}$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 8

Column	Representative term structure
T1	$+\frac{\lambda_{12}^{(A)2}}{4}$
T2	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$
T3	$8\lambda_7(\lambda_7)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	$8\lambda_7(\lambda_7)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	$8\lambda_7(\lambda_7)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{\lambda_{12}^{(A)2}}{4}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 8

Column	Representative term structure			
T1	$+8\lambda_9\left(\lambda_9\right)^*$			
T2	$+24\lambda_{11}\left(\lambda_{11}\right)^*$			
T3	$21g_2^4$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_9\left(\lambda_9\right)^*$	$+24\lambda_{11}\left(\lambda_{11}\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_9\left(\lambda_9\right)^*$	$+24\lambda_{11}\left(\lambda_{11}\right)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$21g_2^4$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$21g_2^4$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$21g_2^4$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$21g_2^4$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$21g_2^4$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$21g_2^4$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$21g_2^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 8

Column	Representative term structure	
T1	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	
T2	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	
T3	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	$+g_1^2\left(48g_2^2-48\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 8

Column	Representative term structure	
T1	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	
T2	$+26\lambda_{S_1}^{(A)2}$	
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 8

Column	Representative term structure	
T1	$+6\lambda_{12}^{(A)2}$	
T2	$+6\lambda_{12}^{(\times)2}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

Terms 1 of 8

Column		Representative term structure
T1		$21g_2^4$
T2		$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$
T3		$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$21g_2^4$	—	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$21g_2^4$	—	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$21g_2^4$	$+g_1^2\left(48g_2^2-48\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 8

Column	Representative term structure
T1	$+12g_1^4$
T2	$+2\lambda_{12}^{(1)2}$
T3	$+4\lambda_{HS_2}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+12g_1^4$	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+12g_1^4$	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	$6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$192g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+192g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 8

Column		Representative term structure
T1		$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$
T2		$+14\lambda_{S_2}^{(1)^2}$
T3		$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 8

Column	Representative term structure			
T1	$+26\lambda_{S_2}^{(A)2}$			
T2	$\frac{9g_2^4}{4}$			
T3	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+26\lambda_{S_2}^{(A)2}$	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 8

Column		Representative term structure
T1	$-9g_2^2\lambda_{S_2}^{(1)}$	
T2	$+\frac{3g_1^4}{4}$	
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 8

Column	Representative term structure	
T1	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	
T2	$8\lambda_8\left(\lambda_8\right)^*$	
T3	$+8\lambda_9\left(\lambda_9\right)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 8

Column	Representative term structure	
T1	$+24\lambda_{10}(\lambda_{10})^*$	
T2	$-12g_1^2\lambda_{S_2}^{(1)}$	
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+24\lambda_{10}(\lambda_{10})^*$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+24\lambda_{10}(\lambda_{10})^*$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-12g_1^2\lambda_{S_2}^{(1)}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-48g_1^2\lambda_{S_2}^{(1)}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 8

Column	Representative term structure	
T1	$+6\lambda_{12}^{(A)2}$	
T2	$+6\lambda_{12}^{(\times)2}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 7

Column		Representative term structure
T1	$\lambda_{HS_1}^{(1)}$	$(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+12\lambda_5\ (\lambda_5)^*$	
T3	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+12\lambda_5\ (\lambda_5)^*$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\lambda_{HS_1}^{(1)}(6\text{ Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{ Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{ Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_5\ (\lambda_5)^*$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+4\lambda_5\ (\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 7

Column		Representative term structure
T1		$+12g_1^4$
T2		$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3		$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A-m2	-2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A-p0	0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(1, -1)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(3, -1)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	—	—	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-m2	-2	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-p0	0	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-p2	2	$(3, 1)$	$(1, 2)$	$(2, \frac{3}{2})$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-m4	-4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$	—
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	—
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	—
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 7

Column		Representative term structure
T1		$+6\lambda_1\lambda_{HS_1}^{(1)}$
T2		$+4\lambda_{HS_1}^{(1)2}$
T3		$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 7

Column	Representative term structure
T1	$+\frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3	$+\frac{27g_1^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{27g_1^4}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{27g_1^4}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 5 of 7

Column	Representative term structure
T1	$-15g_1^2\lambda_{HS_1}^{(1)}$
T2	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$
T3	$+24\lambda_7(\lambda_7)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-15g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	$+24\lambda_7(\lambda_7)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-3g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-15g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	$+24\lambda_7(\lambda_7)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-3g_1^2\lambda_{HS_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A)2}}{4}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 6 of 7

Column	Representative term structure
T1	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*)$
T3	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*)$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(1)}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right))$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 7 of 7

Column	Representative term structure	
T1	$+6g_2^4$	
T2	$+6\lambda_{HS_1}^{(A)^2}$	
T3	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 7

Column		Representative term structure
T1	$\lambda_{HS_2}^{(1)}$	$(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$
T2	$+4\lambda_5\left(\lambda_5\right)^*$	
T3	$+6g_2^4$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2,-\frac{3}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2,-\frac{1}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-A-p0	0	(1, 0)	(3, 1)	$(2,\frac{1}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1, -1)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1, 0)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1, 1)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3, -1)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3, 0)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3, 1)	—	$+6\lambda_5\left(\lambda_5\right)^*$	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2,-\frac{1}{2})$	—	$+12\lambda_5\left(\lambda_5\right)^*$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2,\frac{1}{2})$	—	$+12\lambda_5\left(\lambda_5\right)^*$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2,\frac{3}{2})$	—	$+12\lambda_5\left(\lambda_5\right)^*$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2,-\frac{3}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2,-\frac{1}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-E-p0	0	(3, 0)	(3, 1)	$(2,\frac{1}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$
T3-E-p2	2	(3, 1)	(3, 2)	$(2,\frac{3}{2})$	$\lambda_{HS_2}^{(1)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 7

Column		Representative term structure
T1	-	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T2	+	$3g_1^4$
T3	-	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 7

Column		Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$	
T3	$+4\lambda_{HS_2}^{(1)^2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 7

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+6\lambda_{HS_2}^{(A)}{}^2$
T3	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right))$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)}{}^2$	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 7

Column	Representative term structure
T1	$+\frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3	$+\frac{3g_1^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{27g_1^4}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3g_1^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{27g_1^4}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 6 of 7

Column	Representative term structure
T1	$-3g_1^2\lambda_{HS_2}^{(1)}$
T2	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-15g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-15g_1^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 7 of 7

Column		Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 3 \left(\lambda_{10}\right)^*)$	
T2	$+24\lambda_8 \left(\lambda_8\right)^*$	
T3	$+\lambda_{HS_1}^{(1)} (6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 3 \left(\lambda_{10}\right)^*)$	$+24\lambda_8 \left(\lambda_8\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 3 \left(\lambda_{10}\right)^*)$	$+24\lambda_8 \left(\lambda_8\right)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+\lambda_{HS_1}^{(1)} (6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+\lambda_{HS_1}^{(1)} (6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+\lambda_{HS_1}^{(1)} (6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+\lambda_{HS_1}^{(1)} (6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$$

Terms 1 of 9

Column			Representative term structure
T1	$4\lambda_5\left(\lambda_5\right)^*$		
T2	$-12g_2^2\lambda_{12}^{(1)}$		
T3	$+48g_1^4$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	$+48g_1^4$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$3\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{12}^{(1)}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$4\lambda_5\left(\lambda_5\right)^*$	$-12g_2^2\lambda_{12}^{(1)}$	$+48g_1^4$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+48g_1^4$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+48g_1^4$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-30g_1^2\lambda_{12}^{(1)}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+4\lambda_{12}^{(1)2}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+4\lambda_{12}^{(1)2}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+4\lambda_{12}^{(1)2}$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+4\lambda_{12}^{(1)2}$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 9

Column		Representative term structure
T1		$+\frac{9g_2^4}{4}$
T2		$+\frac{27g_1^4}{4}$
T3		$+\frac{3\lambda_{12}^{(A)2}}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{9g_2^4}{4}$	$+\frac{27g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{9g_2^4}{4}$	$+\frac{3g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{9g_2^4}{4}$	$+\frac{27g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{9g_2^4}{4}$	$+\frac{27g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{9g_2^4}{4}$	$+\frac{3g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{9g_2^4}{4}$	$+\frac{27g_1^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+24\lambda_9\left(\lambda_9\right)^*$
T3	$+9\lambda_{10}\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_9\left(\lambda_9\right)^*$	$+9\lambda_{10}\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*$ $+9\lambda_{11}\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_9\left(\lambda_9\right)^*$	$+9\lambda_{10}\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*$ $+9\lambda_{11}\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$
T2	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T3	$+16g_2^4$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 9

Column	Representative term structure	
T1	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$	
T2	$+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$	
T3	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$	$+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$	$+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$
T2	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$
T3	$+ 16 \lambda_{12}^{(A)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 9

Column		Representative term structure
T1	+	$\frac{32\lambda_{12}^{(\times)2}}{3}$
T2		$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3		$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

$\beta_{\lambda_5}^{(1)}$
Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_5(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{HS_2}^{(A)})$
T2	$-\frac{21g_2^2\lambda_5}{2}$
T3	$-\frac{33g_1^2\lambda_5}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_5(6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_5(6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_5(6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 5

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 5

Column		Representative term structure
T1		$+4\lambda_{HS_2}^{(1)}\lambda_5$
T2		$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3		$-9g_2^2\lambda_5$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—

Quantity: λ_5

Terms 4 of 5

Column	Representative term structure	
T1	$-9g_1^2\lambda_5$	
T2	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$	
T3	$+4\lambda_{11}(\lambda_7)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_1^2\lambda_5$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_5$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$	$+4\lambda_{11}(\lambda_7)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_1^2\lambda_5$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_1^2\lambda_5$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_5$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$	$+4\lambda_{11}(\lambda_7)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_1^2\lambda_5$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_5

Terms 5 of 5

Column		Representative term structure
T1		$+4\left(\lambda_8\right)^*\left((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2		$\lambda_5(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$
T3		$\lambda_5\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\left(\lambda_8\right)^*\left((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\left(\lambda_8\right)^*\left((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\lambda_5(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\lambda_5(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\lambda_5(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\lambda_5\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)\right)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_5\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)\right)$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_5\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)\right)$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_5\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)\right)$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure			
T1	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$			
T2	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$			
T3	$-9g_2^2\lambda_{12}^{(A)}$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{12}^{(A)}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{12}^{(A)}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(16g_2^2-30\lambda_{12}^{(A)}\right)$	$-24g_2^2\lambda_{12}^{(A)}$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	—	$-24g_2^2\lambda_{12}^{(A)}$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	—	$-24g_2^2\lambda_{12}^{(A)}$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(16g_2^2-30\lambda_{12}^{(A)}\right)$	$-24g_2^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 6

Column	Representative term structure	
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 6

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T2	$\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$
T3	$+\frac{4}{3}\lambda_{10}(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	$\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+\frac{4}{3}\lambda_{10}(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*)$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	$\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+\frac{4}{3}\lambda_{10}(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*)$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 6

Column	Representative term structure	
T1	$+\frac{4}{3}\lambda_{11}((\lambda_{10})^{*}+5(\lambda_{11})^{*})$	
T2	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	
T3	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{4}{3}\lambda_{11}((\lambda_{10})^{*}+5(\lambda_{11})^{*})$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{4}{3}\lambda_{11}((\lambda_{10})^{*}+5(\lambda_{11})^{*})$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 6

Column	Representative term structure	
T1	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	
T2	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	
T3	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$-6g_1^2\lambda_{12}^{(A)}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-6g_1^2\lambda_{12}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(\times)}}{d\mu}=\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	
T2	$+\frac{33g_2^4}{4}$	
T3	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 5

Column	Representative term structure	
T1	$+12g_1^4$	
T2	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	
T3	$+\lambda_{12}^{(1)2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12g_1^4$	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+12g_1^4$	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 5

Column	Representative term structure	
T1	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	
T2	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	
T3	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 5

Column		Representative term structure
T1		$+4\lambda_{12}^{(A)2}$
T2		$+11\lambda_{12}^{(\times)2}$
T3		$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 5

Column		Representative term structure
T1		$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T2		$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\lambda_{HS_1}^{(A)}$	$(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T2	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	
T3	$+g_1^2\left(-18g_2^2-15\lambda_{HS_1}^{(A)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,−2)	(3,−1)	(2,− $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,−1)	(3,0)	(2,− $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,− $\frac{3}{2}$)	(2,− $\frac{1}{2}$)	(1,−1)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-18g_2^2-15\lambda_{HS_1}^{(A)}\right)$
T3-B-m1	-1	(2,− $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-C-m3	-3	(2,− $\frac{3}{2}$)	(2,− $\frac{1}{2}$)	(3,−1)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-18g_2^2-15\lambda_{HS_1}^{(A)}\right)$
T3-C-m1	-1	(2,− $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$\lambda_{HS_1}^{(A)}(6\text{ Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{ Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{ Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-D-m2	-2	(3,−1)	(1,0)	(2,− $\frac{1}{2}$)	—	$-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	$-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	$-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-m4	-4	(3,−2)	(3,−1)	(2,− $\frac{3}{2}$)	—	$-\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-m2	-2	(3,−1)	(3,0)	(2,− $\frac{1}{2}$)	—	$-\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	$-\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	$-\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_1}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 6

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+\frac{64}{3}\lambda_7(\lambda_7)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	$+\frac{64}{3}\lambda_7(\lambda_7)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	$+\frac{64}{3}\lambda_7(\lambda_7)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 6

Column	Representative term structure
T1	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)$
T3	$+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	$+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	$+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 5 of 6

Column		Representative term structure
T1	$\lambda_{HS_1}^{(A)}$	$(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$
T2	$+g_1^2$	$\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$
T3	$-\frac{33g_2^2}{2}$	$\lambda_{HS_1}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	$+g_1^2\left(-8g_2^2-\frac{51\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_1}^{(A)}(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S_1}^{(A)})$	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T2	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$
T3	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 7

Column		Representative term structure
T1	$\lambda_{HS_2}^{(A)}$	$(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$
T2	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	
T3	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 7

Column		Representative term structure
T1	-	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$
T2	+	$2\lambda_1\lambda_{HS_2}^{(A)}$
T3	+	$8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
T2	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$
T3	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	$+g_1^2\left(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$
T2	$+16\lambda_5 \left(\lambda_5\right)^*$
T3	$+g_1^2 \left(-6g_2^2 - 3\lambda_{HS_2}^{(A)}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	$+g_1^2 \left(-6g_2^2 - 3\lambda_{HS_2}^{(A)}\right)$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	$+g_1^2 \left(-6g_2^2 - 3\lambda_{HS_2}^{(A)}\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\lambda_{HS_2}^{(A)} (6 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right) + 2 \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right) + 6 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right))$	$+16\lambda_5 \left(\lambda_5\right)^*$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_2}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*)$
T2	$+64\lambda_8\left(\lambda_8\right)^*$
T3	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*)$	$+64\lambda_8\left(\lambda_8\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*)$	$+64\lambda_8\left(\lambda_8\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

$\beta_{\lambda_7}^{(1)}$

Quantity: λ_7

$16\pi^2\mu\frac{d\lambda_7}{d\mu}=\beta_{\lambda_7}^{(1)}$

Terms 1 of 4

Column		Representative term structure
T1	$\lambda_7(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	
T2	$-2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}(\lambda_5)^*$	
T3	$+2\lambda_{11}(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_7(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$-2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}(\lambda_5)^*$	$+2\lambda_{11}(\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_7(6\operatorname{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$	$-2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}(\lambda_5)^*$	$+2\lambda_{11}(\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 2 of 4

Column	Representative term structure	
T1	$-9g_2^2\lambda_7$	
T2	$-3g_1^2\lambda_7$	
T3	$+2\lambda_1\lambda_7$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$	$+2\lambda_1\lambda_7$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$	$+2\lambda_1\lambda_7$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 3 of 4

Column	Representative term structure	
T1	$+8\lambda_{HS_1}^{(1)}\lambda_7$	
T2	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$	
T3	$+2\lambda_7\lambda_{S_1}^{(1)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$	$+2\lambda_7\lambda_{S_1}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$	$+2\lambda_7\lambda_{S_1}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 4 of 4

Column	Representative term structure			
T1	$+4\lambda_8\lambda_9$			

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+4\lambda_8\lambda_9$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+4\lambda_8\lambda_9$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—

$\beta_{\lambda_{12}}^{(1)}$

Quantity: λ_{12}

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 4

Column		Representative term structure
T1	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T2	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	
T3	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure	
T1	$-9g_2{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T2	$-3g_1{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T3	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$-9g_2{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-3g_1{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$-9g_2{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-3g_1{}^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: λ_{12}

Terms 3 of 4

Column	Representative term structure
T1	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$
T3	$+\lambda_{HS_1}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11})$	$+\lambda_{HS_1}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{12}

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$
T2	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$
T3	$-6\lambda_7\lambda_5$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	$-6\lambda_7\lambda_5$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10})$	$+\lambda_{HS_2}^{(A)}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	$-6\lambda_7\lambda_5$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{\lambda_8}^{(1)}$

Quantity: λ_8

$$16\pi^2\mu\frac{d\lambda_8}{d\mu}=\beta_{\lambda_8}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_8(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$
T2	$+4\lambda_7\left(\lambda_9\right)^*$
T3	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_8(6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+4\lambda_7\ (\lambda_9)^*$	$+2\ (\lambda_5)^*\ ((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_8(6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+4\lambda_7\ (\lambda_9)^*$	$+2\ (\lambda_5)^*\ ((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_8

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_8$
T2	$-3g_1^2\lambda_8$
T3	$+2\lambda_1\lambda_8$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$	$+2\lambda_1\lambda_8$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$	$+2\lambda_1\lambda_8$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_8

Terms 3 of 3

Column	Representative term structure	
T1	$+8\lambda_{HS_2}^{(1)}\lambda_8$	
T2	$+4\lambda_{HS_2}^{(A)}\lambda_8$	
T3	$+2\lambda_8\lambda_{S_2}^{(1)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$	$+2\lambda_8\lambda_{S_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1		$\frac{3g_2^4}{4}$
T2	$+g_1^2$	$^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$
T3	$+g_2^2$	$^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 6

Column		Representative term structure
T1		$+\frac{3g_1^4}{2}$
T2		$+\frac{\lambda_{12}^{(1)2}}{4}$
T3		$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 6

Column		Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$	
T2	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	
T3	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 6

Column	Representative term structure	
T1	$+22\lambda_{S_1}^{(A)2}$	
T2	$+g_1^2\left(36g_2^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$	
T3	$+24g_1^4$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+22\lambda_{S_1}^{(A)2}$	$+g_1^2\left(36g_2^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$	$+24g_1^4$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+22\lambda_{S_1}^{(A)2}$	—	—

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 6

Column	Representative term structure		
T1	$+ \frac{3\lambda_{12}^{(1)2}}{4}$		
T2	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$		
T3	$+ 12\lambda_{12}^{(A)2}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12\lambda_{12}^{(A)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12\lambda_{12}^{(A)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12\lambda_{12}^{(A)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12\lambda_{12}^{(A)2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 6 of 6

Column		Representative term structure
T1		$-\frac{\lambda_{12}^{(\times)^2}}{2}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$

$\beta_{\lambda_{11}}^{(1)}$

Quantity: λ_{11}

$$16\pi^2\mu\frac{d\lambda_{11}}{d\mu}=\beta_{\lambda_{11}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure			
T1	$4\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$			
T2	$-9g_2^2\lambda_{11}$			
T3	$-3g_1^2\lambda_{11}$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$4\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$4\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{11}

Terms 2 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: λ_{11}

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$+12\lambda_{S_1}^{(1)}\lambda_{11}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$+12\lambda_{S_1}^{(1)}\lambda_{11}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$+12\lambda_{S_1}^{(1)}\lambda_{11}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

$\beta_{\lambda_9}^{(1)}$

Quantity: λ_9

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu}=\beta_{\lambda_9}^{(1)}$$

Terms 1 of 4

Column	Representative term structure	
T1	$4\lambda_7\left(\lambda_8\right)^*$	
T2	$-9g_2^2\lambda_9$	
T3	$-3g_1^2\lambda_9$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$	$-3g_1^2\lambda_9$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$	$-3g_1^2\lambda_9$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_9$
T2	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 3 of 4

Column	Representative term structure	
T1	$+2\lambda_{S_1}^{(1)}\lambda_9$	
T2	$+2\lambda_9\lambda_{S_2}^{(1)}$	
T3	$+5\lambda_{10}^2$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—	—	—

Quantity: λ_9

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{10}\lambda_{11}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	—
T3-A-p0	0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(1, -1)$	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	$+2\lambda_{10}\lambda_{11}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(3, -1)$	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	$+2\lambda_{10}\lambda_{11}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	—
T3-D-m2	-2	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	—
T3-D-p0	0	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	—
T3-D-p2	2	$(3, 1)$	$(1, 2)$	$(2, \frac{3}{2})$	—
T3-E-m4	-4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	—
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	—
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	—
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	—

$\beta_{\lambda_{10}}^{(1)}$

Quantity: λ_{10}

$16\pi^2\mu\frac{d\lambda_{10}}{d\mu}=\beta_{\lambda_{10}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T2	$+4\lambda_9\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$
T3	$-9g_2^2\lambda_{10}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$+4\lambda_9\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$	$-9g_2^2\lambda_{10}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$+4\lambda_9\left(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*\right)$	$-9g_2^2\lambda_{10}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 2 of 3

Column	Representative term structure
T1	$-3g_1^2\lambda_{10}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1		$\frac{3g_2^4}{4}$
T2	$+g_1^2$	$\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$
T3	$+g_2^2$	$(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 6

Column		Representative term structure
T1		$+\frac{3g_1^4}{2}$
T2		$+\frac{\lambda_{12}^{(1)2}}{4}$
T3		$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 6

Column		Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$	
T2	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	
T3	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+13\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 6

Column	Representative term structure	
T1	$+22\lambda_{S_2}^{(A)^2}$	
T2	$+\frac{3\lambda_{12}^{(1)^2}}{4}$	
T3	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	$+\frac{3\lambda_{12}^{(1)^2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	$+\frac{3\lambda_{12}^{(1)^2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	$+\frac{3\lambda_{12}^{(1)^2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+22\lambda_{S_2}^{(A)^2}$	$+\frac{3\lambda_{12}^{(1)^2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 6

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)2}$
T2	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$	—
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$	—
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$	—
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$+24g_1^4$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+24g_1^4$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

Quantity: 1, 1, 3, 3

$$16\pi^2\,\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure		
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1134}}^{(1)}$

Quantity: 1, 1, 3, 4

$_{16\pi^2}^{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$

Column		Representative term	structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1144}}^{(1)}$

Quantity: 1, 1, 4, 4

$_{16\pi^2}\mu\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$

Column		Representative term	structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1155}}^{(1)}$
Quantity: 1, 1, 5, 5

$$_{16\pi^2}\mu\frac{dC_{1155}}{d\mu}=\beta_{C_{1155}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$	
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$	
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 5

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2		$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3		$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{1156}}^{(1)}$

Quantity: 1, 1, 5, 6

$$_{16\pi^2}\mu\frac{dC_{1156}}{d\mu}=\beta_{C_{1156}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2		$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3		$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 6

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$$_{16\pi^2}\mu\frac{dC_{1157}}{d\mu}=\beta_{C_{1157}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 7

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^4 y e \overline{y_1 y_2 y e}}{M F}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^4 y e \overline{y_1 y_2 y e}}{12 M F}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^4 y e \overline{y_1 y_2 y e}}{12 M F}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^4 y e \overline{y_1 y_2 y e}}{12 M F}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{C_{1158}}^{(1)}$
Quantity: 1, 1, 5, 8

$$_{16\pi^2}\mu\frac{dC_{1158}}{d\mu}=\beta_{C_{1158}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 1, 1, 5, 8

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ie\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}ie\overline{y_1}y_2\overline{y}e}{12MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ie\overline{y_1}y_2\overline{y}e}{12MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ie\overline{y_1}y_2\overline{y}e}{12MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{C_{11511}}^{(1)}$
Quantity: 1, 1, 5, 11

$$_{16\pi^2\mu}\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 1, 1, 5, 11

Terms 2 of 6

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$
T2	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{MF}$
T3	$\frac{g_1^2 y_1 y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	$-\frac{3g_1^2 y_1 y_2}{2MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	$-\frac{3g_1^2 y_1 y_2}{2MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 11

Terms 3 of 6

Column		Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 11

Terms 4 of 6

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	—	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 11

Terms 5 of 6

Column		Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 11

Terms 6 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2\overline{y_e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y_e}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y_e}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2y_e}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{11512}}^{(1)}$
Quantity: 1, 1, 5, 12

$$16\pi^2\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 1, 1, 5, 12

Terms 2 of 6

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ye\overline{y_1y_2ye}}{MF}$
T2	$\sqrt{2} \cdot \frac{3}{4}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3	ig_1^2	$\frac{\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3}{4}\frac{ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{4}\frac{ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{4}\frac{ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	$-\frac{3ig_1^2\overline{y_1y_2}}{2MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	$-\frac{3ig_1^2\overline{y_1y_2}}{2MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{\sqrt{2} \cdot \frac{3}{4}ig_2^2\overline{y_1y_2}}{3MF}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 12

Terms 3 of 6

Column		Representative term structure
T1		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ilambda12In v2\overline{y_1y_2}}{MF}$
T3		$\frac{iy e\overline{y_1y_2}ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12In v2\overline{y_1y_2}}{3MF}$	$\frac{iy e\overline{y_1y_2}ye}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12In v2\overline{y_1y_2}}{3MF}$	$\frac{iy e\overline{y_1y_2}ye}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12In v2\overline{y_1y_2}}{3MF}$	$\frac{iy e\overline{y_1y_2}ye}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 12

Terms 4 of 6

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	—	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 12

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} i \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 5, 12

Terms 6 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\mu\frac{dC_{11513}}{d\mu}=\beta_{C_{11513}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 5, 13

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{11514}}{d\mu}=\beta_{C_{11514}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$

Quantity: 1, 1, 5, 14

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{1166}}{d\mu}=\beta_{C_{1166}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\lambda S1Inv1\overline{y_1}^2}{MF}$
T2		$\frac{g_2^2\overline{y_1}^2}{MF}$
T3		$\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\lambda S1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 6

Terms 2 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$	
T2	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$	
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{1167}}^{(1)}$
Quantity: 1, 1, 6, 7

$$_{16\pi^2}\mu\frac{dC_{1167}}{d\mu}=\beta_{C_{1167}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 7

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^4 i y e \overline{y_1} y_2 y \overline{e}}{12 M F}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^4 i y e \overline{y_1} y_2 y \overline{e}}{12 M F}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^4 i y e \overline{y_1} y_2 y \overline{e}}{12 M F}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^4 i y e \overline{y_1} y_2 y \overline{e}}{12 M F}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{C_{1168}}^{(1)}$

Quantity: 1, 1, 6, 8

$$_{16\pi^2}\mu\frac{dC_{1168}}{d\mu}=\beta_{C_{1168}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 8

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3}{4}\frac{ye\overline{y_1y_2y\overline{e}}}{12MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3}{4}\frac{ye\overline{y_1y_2y\overline{e}}}{12MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3}{4}\frac{ye\overline{y_1y_2y\overline{e}}}{12MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{C_{11611}}^{(1)}$
Quantity: 1, 1, 6, 11

$$_{16\pi^2\mu}\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12Inv1}\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 11

Terms 2 of 6

Column		Representative term structure
T1	$\frac{3}{3^4} \frac{ye\overline{y_1y_2ye}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{MF}$	
T3	$\frac{ig_1^2 \overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}} ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	$\frac{3ig_1^2 \overline{y_1y_2}}{2MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	$\frac{3ig_1^2 \overline{y_1y_2}}{2MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1y_2}}{3MF}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 11

Terms 3 of 6

Column		Representative term structure
T1		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{iy_e\overline{y_1y_2}y\overline{e}}{MF}$
T3		$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy_e\overline{y_1y_2}y\overline{e}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy_e\overline{y_1y_2}y\overline{e}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy_e\overline{y_1y_2}y\overline{e}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 11

Terms 4 of 6

Column	Representative term structure
T1	$\frac{ilambda12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{ilambda12Inv2y_1y_2}{3MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{ilambda12Inv2y_1y_2}{3MF}$	—	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{ilambda12Inv2y_1y_2}{3MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 11

Terms 5 of 6

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} \overline{y_2} \overline{y} e}{M F}$
T2		$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1} \overline{y_2}}{M F}$
T3		$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1} \overline{y_2}}{M F}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3} i y e \overline{y_1} \overline{y_2} \overline{y} e}{12 M F}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3} i y e \overline{y_1} \overline{y_2} \overline{y} e}{12 M F}$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3} i y e \overline{y_1} \overline{y_2} \overline{y} e}{12 M F}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1} \overline{y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1} \overline{y_2}}{8 M F}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1} \overline{y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1} \overline{y_2}}{8 M F}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1} \overline{y_2}}{6 M F}$	$\frac{9 \cdot 3^{\frac{3}{4}} i g_1^2 \overline{y_1} \overline{y_2}}{8 M F}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 11

Terms 6 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{11612}}^{(1)}$
Quantity: 1, 1, 6, 12

$$16\pi^2\,\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 12

Terms 2 of 6

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$
T2	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{MF}$
T3	$\frac{g_1^2 y_1 y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	$-\frac{3 g_1^2 y_1 y_2}{2MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	$-\frac{3 g_1^2 y_1 y_2}{2MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 12

Terms 3 of 6

Column		Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3} lambda 12 Inv 2 \overline{y_1 y_2}}{3MF}$	$\frac{ye \overline{y_1 y_2 y \overline{e}}}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 12

Terms 4 of 6

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	—	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 12

Terms 5 of 6

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1}y_2}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 6, 12

Terms 6 of 6

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2\overline{y_e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu\frac{dC_{11613}}{d\mu}=\beta_{C_{11613}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 6, 13

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$

$\beta_{C_{11614}}^{(1)}$
Quantity: 1, 1, 6, 14

$$16\pi^2\,\mu\frac{dC_{11614}}{d\mu}=\beta_{C_{11614}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 6, 14

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{6MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11711}}^{(1)}$
Quantity: 1, 1, 7, 11

$$_{16\pi^2}\mu\frac{dC_{11711}}{d\mu}=\beta_{C_{11711}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 7, 11

Terms 2 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$
T3		$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$

Quantity: 1, 1, 7, 11

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$_{16\pi^2}\mu\frac{dC_{11712}}{d\mu}=\beta_{C_{11712}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 7, 12

Terms 2 of 3

Column		Representative term structure
T1		$\frac{\sqrt{11}i\lambda da12Inv2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Quantity: 1, 1, 7, 12

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{11715}}{d\mu}=\beta_{C_{11715}}^{(1)}$$

Terms 1 of 2

	Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 7, 15

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{11716}}{d\mu}=\beta_{C_{11716}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{11811}}{d\mu}=\beta_{C_{11811}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Quantity: 1, 1, 8, 11

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$	
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 8, 11

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{11812}}{d\mu}=\beta_{C_{11812}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 8, 12

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$	
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$

Quantity: 1, 1, 8, 12

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$_{16\pi^2}\mu\frac{dC_{11815}}{d\mu}=\beta_{C_{11815}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 8, 15

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3		$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{11816}}{d\mu}=\beta_{C_{11816}}^{(1)}$$

Terms 1 of 2

	Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$$16\pi^2\mu\frac{dC_{11911}}{d\mu}=\beta_{C_{11911}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 9, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ie\overline{y_1y_2y\epsilon}}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3}{4}\frac{ie\overline{y_1y_2y\epsilon}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3}{4}\frac{ie\overline{y_1y_2y\epsilon}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3}{4}\frac{ie\overline{y_1y_2y\epsilon}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$$16\pi^2\mu\frac{dC_{11912}}{d\mu}=\beta_{C_{11912}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{\text{lambda12Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 9, 12

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^4}{12MF}ye\overline{y_1y_2y\overline{e}}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^4}{12MF}ye\overline{y_1y_2y\overline{e}}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^4}{12MF}ye\overline{y_1y_2y\overline{e}}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3^4}{12MF}ye\overline{y_1y_2y\overline{e}}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$$16\pi^2\,\mu\frac{dC_{11913}}{d\mu}=\beta_{C_{11913}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 9, 13

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 9, 13

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{11914}}{d\mu}=\beta_{C_{11914}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 9, 14

Terms 2 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 9, 14

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$

$$16\pi^2\,_{\mu}\frac{dC_{111011}}{d\mu}=\beta_{C_{111011}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}g_2^2\overline{y_1y_2}$	MF
T2	$\frac{3}{4}\lambda\overline{Inv1y_1y_2}$	MF
T3	$\frac{3}{4}g_1^2\overline{y_1y_2}$	MF

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 10, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$$_{16\pi^2\mu}\frac{dC_{111012}}{d\mu}=\beta_{C_{111012}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 10, 12

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{3\,4} iye\overline{y_1y_2ye}$	MF

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} iye\overline{y_1y_2ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} iye\overline{y_1y_2ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}} iye\overline{y_1y_2ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—

$\beta_{C_{111013}}^{(1)}$
Quantity: 1, 1, 10, 13

$$16\pi^2\,{}_{\mu}\frac{dC_{111013}}{d\mu}=\beta_{C_{111013}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 10, 13

Terms 2 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$
T2		$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{MF}$
T3		$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

Quantity: 1, 1, 10, 13

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$$16\pi^2\,\mu\frac{dC_{111014}}{d\mu}=\beta_{C_{111014}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 10, 14

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 10, 14

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$\beta_{C_{111111}}^{(1)}$
Quantity: 1, 1, 11, 11

$$_{16\pi^2\mu}\frac{dC_{111111}}{d\mu}=\beta_{C_{111111}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 11, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$	
T2	$\frac{\sqrt{3}\lambda daS2Inv1\overline{y_2}^2}{MF}$	
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\bar{e}}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}\lambda daS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\bar{e}}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{111112}}^{(1)}$
Quantity: 1, 1, 11, 12

$$16\pi^2\mu\frac{dC_{111112}}{d\mu}=\beta_{C_{111112}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Invl\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Invl\overline{y_2}^2}{2MF}$	$\frac{iye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 11, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{111212}}^{(1)}$
Quantity: 1, 1, 12, 12

$$_{16\pi^2}\mu\frac{dC_{111212}}{d\mu}=\beta_{C_{111212}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\lambda S2Inv1\overline{y_2}^2}{MF}$
T2		$\frac{g_2^2\overline{y_2}^2}{MF}$
T3		$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\lambda S2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 1, 12, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y}_2^2}{MF}$
T2	$\frac{\sqrt{3}\lambda S2Inv1\overline{y}_2^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}\overline{e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y}_2^2}{4MF}$	$\frac{\sqrt{3}\lambda S2Inv1\overline{y}_2^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}\overline{e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{1213}}^{(1)}$

Quantity: 1, 2, 1, 3

$_{16\pi^2}\mu\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$

Column		Representative term	structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1214}}^{(1)}$

Quantity: 1, 2, 1, 4

$_{16\pi^2}\mu\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$

Column	Representative term	structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$	
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1223}}^{(1)}$

Quantity: 1, 2, 2, 3

$_{16\pi^2} \mu \frac{dC_{1223}}{d\mu} = \beta_{C_{1223}}^{(1)}$

Column	Representative term	structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$	
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1224}}^{(1)}$

Quantity: 1, 2, 2, 4

$_{16\pi^2}^{\mu}\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$

Column		Representative term	structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$$_{16\pi^2}\mu\frac{dC_{1257}}{d\mu}=\beta_{C_{1257}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{g_2^2\overline{y_1}^2}{MF}$
T2		$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3		$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$-\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{1258}}^{(1)}$
Quantity: 1, 2, 5, 8

$$_{16\pi^2}\mu\frac{dC_{1258}}{d\mu}=\beta_{C_{1258}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{1259}}^{(1)}$
Quantity: 1, 2, 5, 9

$$_{16\pi^2}\mu\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 9

Terms 2 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$	
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 9

Terms 3 of 5

Column		Representative term structure
T1	$\frac{g_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{g_2^2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{ye \overline{y_1 y_2 y e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 9

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{MF}$
T2	$\frac{lambda12Inv2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}g_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{6MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{6MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}lambda12Inv2y_1y_2}{6MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{lambda12Inv2y_1y_2}{6MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{lambda12Inv2y_1y_2}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{lambda12Inv2y_1y_2}{6MF}$	$-\frac{\sqrt{3}g_1^2y_1y_2}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 9

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12510}}^{(1)}$
Quantity: 1, 2, 5, 10

$$16\pi^2\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 10

Terms 2 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 10

Terms 3 of 5

Column		Representative term structure
T1		$\frac{ig_1^2 \overline{y_1 y_2}}{MF}$
T2		$\frac{ig_2^2 \overline{y_1 y_2}}{MF}$
T3		$\frac{iy e \overline{y_1 y_2 y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3ig_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3ig_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 10

Terms 4 of 5

Column	Representative term structure	
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 5, 10

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12511}}^{(1)}$
Quantity: 1, 2, 5, 11

$$16\pi^2\,\mu\frac{dC_{12511}}{d\mu}=\beta_{C_{12511}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12512}}^{(1)}$
Quantity: 1, 2, 5, 12

$$16\pi^2\,\mu\frac{dC_{12512}}{d\mu}=\beta_{C_{12512}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$$_{16\pi^2}\mu\frac{dC_{12515}}{d\mu}=\beta_{C_{12515}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 5, 15

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}i\lambda da12Inv2y_1y_2}{MF}$	
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$	
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$

Quantity: 1, 2, 5, 15

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{12516}}{d\mu}=\beta_{C_{12516}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 2, 5, 16

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$	
T2	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{MF}$	
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{6MF}$

Quantity: 1, 2, 5, 16

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{1267}}{d\mu}=\beta_{C_{1267}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$$_{16\pi^2}\mu\frac{dC_{1268}}{d\mu}=\beta_{C_{1268}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\lambda S1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\lambda S1Inv1\overline{y_1}^2}{4MF}$	$\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{ye\overline{y_1}^2\overline{y_e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 8

Terms 2 of 2

Column		Representative term structure
T1	MF	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2		$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3		$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$$_{16\pi^2}\mu\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 1, 2, 6, 9

Terms 2 of 5

Column		Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 9

Terms 3 of 5

Column		Representative term structure
T1	$\frac{ig_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{ig_2^2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{\sqrt{3} i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3ig_1^2 \overline{y_1 y_2}}{4MF}$	$-\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3} i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{6MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3} i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{6MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3ig_1^2 \overline{y_1 y_2}}{4MF}$	$-\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3} i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{6MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 9

Terms 4 of 5

Column	Representative term structure
T1	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{MF}$
T2	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{MF}$
T3	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{6MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{ilambda 12 I n v 2 \overline{y_1} y_2}{6MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 9

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12610}}^{(1)}$
Quantity: 1, 2, 6, 10

$$16\pi^2\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 10

Terms 2 of 5

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$	
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 10

Terms 3 of 5

Column		Representative term structure
T1		$\frac{g_1^2 \overline{y_1 y_2}}{MF}$
T2		$\frac{g_2^2 \overline{y_1 y_2}}{MF}$
T3		$\frac{ye \overline{y_1 y_2 y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$	$\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{ye \overline{y_1 y_2 y e}}{8MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 10

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 6, 10

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12611}}^{(1)}$
Quantity: 1, 2, 6, 11

$$16\pi^2\,\mu\frac{dC_{12611}}{d\mu}=\beta_{C_{12611}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12612}}^{(1)}$
Quantity: 1, 2, 6, 12

$$16\pi^2\,\mu\frac{dC_{12612}}{d\mu}=\beta_{C_{12612}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$_{16\pi^2}\mu\frac{dC_{12615}}{d\mu}=\beta_{C_{12615}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{6MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 2, 6, 15

Terms 2 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$

Quantity: 1, 2, 6, 15

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{6MF}$

$$_{16\pi^2}\mu\frac{dC_{12616}}{d\mu}=\beta_{C_{12616}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}i\lambda da12Inv2y_1y_2}{MF}$	
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$	
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$

Quantity: 1, 2, 6, 16

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi}{}^2{}_{\mu}\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 11

Terms 2 of 5

Column	Representative term structure
T1	$\frac{ye\overline{y_1y_2ye}}{MF}$
T2	$\frac{\lambda 12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{\lambda 12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{\lambda 12Inv2\overline{y_1y_2}}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{\lambda 12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 3 of 5

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 11

Terms 4 of 5

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{g_1^2 y_1 y_2}{MF}$
T2	$\frac{3}{4}$	$\frac{g_2^2 y_1 y_2}{MF}$
T3	$\frac{3}{4}$	$\frac{ye y_1 y_2 y e}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 11

Terms 5 of 5

Column		Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\,\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3ig_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3ig_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{MF}$
T2	$\frac{ilambda 12Inv 2 \overline{y_1} y_2}{MF}$
T3	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{8MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{ilambda 12Inv 2 \overline{y_1} y_2}{6MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{ilambda 12Inv 2 \overline{y_1} y_2}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{ilambda 12Inv 2 \overline{y_1} y_2}{6MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 12

Terms 3 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$
T3		$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 12

Terms 4 of 5

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9 \cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 7, 12

Terms 5 of 5

Column		Representative term structure
T1	$\frac{3}{4} i y e \overline{y_1 y_2 y e}$	$M F$
T2	$\sqrt{2} \cdot \frac{3}{4} i g_2^2 \overline{y_1 y_2}$	$M F$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12713}}^{(1)}$

Quantity: 1, 2, 7, 13

$16\pi^2\mu\frac{dC_{12713}}{d\mu}=\beta_{C_{12713}}^{(1)}$

Column Representative term structure	
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12714}}^{(1)}$

Quantity: 1, 2, 7, 14

$16\pi^2\mu\frac{dC_{12714}}{d\mu}=\beta_{C_{12714}}^{(1)}$

Column Representative term structure	
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$$_{16\pi^2}\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{iy e\overline{y_1y_2y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3ig_1^2\overline{y_1y_2}}{4MF}$	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{iy e\overline{y_1y_2y e}}{8MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{iy e\overline{y_1y_2y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3ig_1^2\overline{y_1y_2}}{4MF}$	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{iy e\overline{y_1y_2y e}}{8MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 11

Terms 2 of 5

Column	Representative term structure		
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$		
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 11

Terms 3 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{24MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2y_1y_2}{24MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2y_1y_2}{24MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2y_1y_2}{24MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 11

Terms 4 of 5

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\sqrt{2} \cdot \frac{3}{4}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{9 \cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 11

Terms 5 of 5

Column		Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$	
T2	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\lambda da12Inv2\overline{y_1y_2}}{6MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\lambda da12Inv2\overline{y_1y_2}}{6MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\lambda da12Inv2\overline{y_1y_2}}{6MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 12

Terms 2 of 5

Column		Representative term structure
T1	$\frac{ye\overline{y_1y_2ye}}{MF}$	
T2	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{ye\overline{y_1y_2ye}}{8MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{lambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 12

Terms 3 of 5

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 12

Terms 4 of 5

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{g_1^2 y_1 y_2}{MF}$
T2	$\frac{3}{4}$	$\frac{g_2^2 y_1 y_2}{MF}$
T3	$\frac{3}{4}$	$\frac{ye y_1 y_2 ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 8, 12

Terms 5 of 5

Column		Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{12813}}^{(1)}$
Quantity: 1, 2, 8, 13

$$16\pi^2\,\mu\frac{dC_{12813}}{d\mu}=\beta_{C_{12813}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12814}}^{(1)}$
Quantity: 1, 2, 8, 14

$$16\pi^2\,\mu\frac{dC_{12814}}{d\mu}=\beta_{C_{12814}}^{(1)}$$

Column Representative term structure	
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$_{16\pi^2}\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{\lambda S2Inv1\overline{y_2}^2}{MF}$
T2		$\frac{g_2^2\overline{y_2}^2}{MF}$
T3		$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\lambda S2Inv1\overline{y_2}^2}{4MF}$	$\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 9, 11

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\textit{ambda}S2\textit{Inv}1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}\lambda\textit{ambda}S2\textit{Inv}1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 3 of 5

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$

Quantity: 1, 2, 9, 11

Terms 4 of 5

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 9, 11

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$16\pi^2\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iy e\overline{y_2}^2\overline{y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iy e\overline{y_2}^2\overline{y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 9, 12

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 9, 12

Terms 3 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 4 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 9, 12

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}ye\overline{y_1}y_2y\overline{e}}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$\frac{ilambda S2Inv1\overline{y_2}^2}{MF}$
T2		$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3		$\frac{iy e\overline{y_2}^2\overline{y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambda S2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iy e\overline{y_2}^2\overline{y e}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 10, 11

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 1, 2, 10, 11

Terms 3 of 5

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}\lambda a_{12}Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda a_{12}Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 2, 10, 11

Terms 4 of 5

Column		Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$	
T2		
T3		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$\frac{\sqrt{6i}g_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$\frac{\sqrt{6i}g_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$\frac{\sqrt{6i}g_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}\lambda\overline{Inv1\overline{y_1y_2}}}{6MF}$	$\frac{\sqrt{6i}g_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 10, 11

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$$_{16\pi^2}\mu\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 2 of 5

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$	
T2	$\frac{\sqrt{3}\lambda S2Inv1\overline{y_2}^2}{MF}$	
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$-\frac{\sqrt{3}\lambda S2Inv1\overline{y_2}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{24MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 3 of 5

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2y_1y_2}{6MF}$	$-\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$

Quantity: 1, 2, 10, 12

Terms 4 of 5

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 10, 12

Terms 5 of 5

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$\beta_{C_{2211}}^{(1)}$

Quantity: 2, 2, 1, 1

$_{16\pi^2}^{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$

Column		Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$	
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2212}}^{(1)}$

Quantity: 2, 2, 1, 2

$_{16\pi^2}\mu\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$

Column	Representative term	structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$	
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{2\sqrt{6}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2222}}^{(1)}$

Quantity: 2, 2, 2, 2

$_{16\pi^2}\mu\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$

Column		Representative term	structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$		
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2257}}^{(1)}$
Quantity: 2, 2, 5, 7

$$_{16\pi^2}\mu\frac{dC_{2257}}{d\mu}=\beta_{C_{2257}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 5, 7

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu\frac{dC_{2258}}{d\mu}=\beta_{C_{2258}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 5, 8

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{ie\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\sqrt{2} \cdot \frac{3}{4}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}}ie\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ie\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}}ie\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\,\mu\frac{dC_{22511}}{d\mu}=\beta_{C_{22511}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 2, 2, 5, 11

Terms 2 of 3

Column	Representative term structure		
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$		
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$		
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 5, 11

Terms 3 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\mu\frac{dC_{22512}}{d\mu}=\beta_{C_{22512}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 5, 12

Terms 2 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2\overline{y_1 y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 2, 2, 5, 12

Terms 3 of 3

Column		Representative term structure
T1	$\frac{3}{4} \frac{ig_2^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3}{4} \frac{iy e \overline{y_1 y_2 y e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} iy e \overline{y_1 y_2 y e}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} iy e \overline{y_1 y_2 y e}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} iy e \overline{y_1 y_2 y e}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$16\pi^2\mu\frac{dC_{22513}}{d\mu}=\beta_{C_{22513}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 2, 2, 5, 13

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 5, 13

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22514}}{d\mu}=\beta_{C_{22514}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$

Quantity: 2, 2, 5, 14

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{2267}}{d\mu}=\beta_{C_{2267}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 6, 7

Terms 2 of 2

Column		Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{M F}$	
T2	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{M F}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12 M F}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12 M F}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e y_1 y_2 y e}{12 M F}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{2268}}^{(1)}$
Quantity: 2, 2, 6, 8

$$_{16\pi^2}\mu\frac{dC_{2268}}{d\mu}=\beta_{C_{2268}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 6, 8

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{22611}}^{(1)}$

Quantity: 2, 2, 6, 11

$$_{16\pi^2}\mu^2\frac{dC_{22611}}{d\mu}=\beta_{C_{22611}}^{(1)}$$

Terms 1 of 3

	Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 6, 11

Terms 2 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}} ilambda 12Inv1\overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 y e}}{12MF}$	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	$-\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} ilambda 12Inv1\overline{y_1 y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	$-\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} ilambda 12Inv1\overline{y_1 y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	$-\frac{11 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} ilambda 12Inv1\overline{y_1 y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: 2, 2, 6, 11

Terms 3 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$16\pi^2\mu\frac{dC_{22612}}{d\mu}=\beta_{C_{22612}}^{(1)}$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 6, 12

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 6, 12

Terms 3 of 3

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$_{16\pi^2}\mu\frac{dC_{22613}}{d\mu}=\beta_{C_{22613}}^{(1)}$

Terms 1 of 3

Column Representative term structure	
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}i\lambda da12Inv2y_1y_2}{MF}$	
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$	
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3\sqrt{11}i\lambda da12Inv2y_1y_2}{22MF}$	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Quantity: 2, 2, 6, 13

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\mu\frac{dC_{22614}}{d\mu}=\beta_{C_{22614}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$\beta_{C_{2277}}^{(1)}$

Quantity: 2, 2, 7, 7

$_{16\pi^2}\mu\frac{dC_{2277}}{d\mu}=\beta_{C_{2277}}^{(1)}$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$	
T2	$\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{MF}$	
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 7, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y\bar{e}}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{2278}}^{(1)}$

Quantity: 2, 2, 7, 8

$_{16\pi^2}\mu\frac{dC_{2278}}{d\mu}=\beta_{C_{2278}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 7, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$_{16\pi^2\mu}\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3g_1^2\overline{y_1y_2}}{2MF}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3g_1^2\overline{y_1y_2}}{2MF}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 7, 9

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 7, 9

Terms 3 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{iy e\overline{y_1y_2y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3ig_1^2\overline{y_1y_2}}{2MF}$	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy e\overline{y_1y_2y e}}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy e\overline{y_1y_2y e}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3ig_1^2\overline{y_1y_2}}{2MF}$	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy e\overline{y_1y_2y e}}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 7, 10

Terms 2 of 3

Column	Representative term structure		
T1	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$		
T2	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$		
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu^2\frac{dC_{22711}}{d\mu}=\beta_{C_{22711}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 7, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$	
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$	
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$

$_{16\pi^2}\mu\frac{dC_{22712}}{d\mu}=\beta_{C_{22712}}^{(1)}$

Terms 1 of 2

	Column	Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22715}}{d\mu}=\beta_{C_{22715}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$

Quantity: 2, 2, 7, 15

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{22716}}{d\mu}=\beta_{C_{22716}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$

Quantity: 2, 2, 7, 16

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$_{16\pi^2}\mu\frac{dC_{2288}}{d\mu}=\beta_{C_{2288}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\lambda S1Inv1\overline{y_1}^2}{MF}$
T2		$\frac{g_2^2\overline{y_1}^2}{MF}$
T3		$\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\lambda S1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 8, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$_{16\pi^2}\mu\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$

Terms 1 of 3

	Column	Representative term structure
T1		$\frac{ig_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3ig_1^2\overline{y_1y_2}}{2MF}$	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3ig_1^2\overline{y_1y_2}}{2MF}$	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 8, 9

Terms 2 of 3

Column	Representative term structure
T1	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{MF}$
T2	$\frac{i \lambda b d a 12 I n v 2 \overline{y_1} y_2}{MF}$
T3	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{4MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{4MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{iy e \overline{y_1} y_2 y \overline{e}}{4MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{i \lambda b d a 12 I n v 2 \overline{y_1} y_2}{3MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{2MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$\frac{i \lambda b d a 12 I n v 2 \overline{y_1} y_2}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{i \lambda b d a 12 I n v 2 \overline{y_1} y_2}{3MF}$	$\frac{\sqrt{3} i g_1^2 \overline{y_1} y_2}{2MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 8, 9

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{g_1^2\overline{y_1y_2}}{MF}$
T2		$\frac{g_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{3g_1^2\overline{y_1y_2}}{2MF}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{3g_1^2\overline{y_1y_2}}{2MF}$	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 8, 10

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{3MF}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{lambda12Inv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 8, 10

Terms 3 of 3

Column		Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\mu\frac{dC_{22811}}{d\mu}=\beta_{C_{22811}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 8, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$$_{16\pi^2}\mu^2\frac{dC_{22812}}{d\mu}=\beta_{C_{22812}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 8, 12

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3		$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$

$$16\pi^2\mu\frac{dC_{22815}}{d\mu}=\beta_{C_{22815}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 2, 2, 8, 15

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$	
T2	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{MF}$	
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$	$\frac{\sqrt{3}lambda12In\nu1\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{MF}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}ye\overline{y_1}y_2ye}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22816}}{d\mu}=\beta_{C_{22816}}^{(1)}$$

Terms 1 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$	$-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$

Quantity: 2, 2, 8, 16

Terms 2 of 3

Column		Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column		Representative term structure
T1		$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{M F}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{2299}}{d\mu}=\beta_{C_{2299}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{g_2^2\overline{y_2}^2}{MF}$
T2		$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3		$\frac{ye\overline{y_2}^2\overline{y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{y\bar{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 9, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\bar{e}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\bar{e}}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$$_{16\pi^2}\mu\frac{dC_{22910}}{d\mu}=\beta_{C_{22910}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2		$\frac{ilambda da S2Inv1\overline{y_2}^2}{MF}$
T3		$\frac{iy e\overline{y_2}^2\overline{y e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambda da S2Inv1\overline{y_2}^2}{2MF}$	$\frac{iy e\overline{y_2}^2\overline{y e}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 9, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$$16\pi^2\mu\frac{dC_{22911}}{d\mu}=\beta_{C_{22911}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Invl\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Invl\overline{y_1y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Invl\overline{y_1y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Invl\overline{y_1y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 9, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$16\pi^2\mu\frac{dC_{22912}}{d\mu}=\beta_{C_{22912}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{\text{lambda12Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}}\text{lambda12Inv1}\overline{y_1y_2}}{6MF}$	$\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 9, 12

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$$_{16\pi^2}\mu^2\frac{dC_{22913}}{d\mu}=\beta_{C_{22913}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda bd a_{12}Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda bd a_{12}Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bd a_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda bd a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bd a_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda bd a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bd a_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda bd a_{12}Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda bd a_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda bd a_{12}Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$

$$16\pi^2\mu\frac{dC_{22914}}{d\mu}=\beta_{C_{22914}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3		$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{221010}}^{(1)}$
Quantity: 2, 2, 10, 10

$$16\pi^2\,_{\mu}\frac{dC_{221010}}{d\mu}=\beta_{C_{221010}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$	
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$	
T3	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 10, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y}_2^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\textit{ambda}S2\textit{Inv}1\overline{y}_2^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}\overline{e}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{\sqrt{3}g_2^2\overline{y}_2^2}{4MF}$	$\frac{\sqrt{3}\lambda\textit{ambda}S2\textit{Inv}1\overline{y}_2^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}\overline{e}}{12MF}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{C_{221011}}^{(1)}$
Quantity: 2, 2, 10, 11

$$16\pi^2{}_{\mu}\frac{dC_{221011}}{d\mu}=\beta_{C_{221011}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3}{4}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3}{4}$	$\frac{\lambda\overline{Inv_1y_1y_2}}{MF}$
T3	$\frac{3}{4}$	$\frac{g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv_1\overline{y_1y_2}}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv_1\overline{y_1y_2}}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv_1\overline{y_1y_2}}}{6MF}$	$-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 10, 11

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{MF}$	
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y e}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{221012}}^{(1)}$
Quantity: 2, 2, 10, 12

$$16\pi^2\,_{\mu}\frac{dC_{221012}}{d\mu}=\beta_{C_{221012}}^{(1)}$$

Terms 1 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$	
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$	
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: 2, 2, 10, 12

Terms 2 of 2

Column		Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$	
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{C_{221013}}^{(1)}$
Quantity: 2, 2, 10, 13

$$_{16\pi^2\mu}\frac{dC_{221013}}{d\mu}=\beta_{C_{221013}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2		$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$
T3		$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{6MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{6MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{6MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{3ig_2^2\overline{y_1y_2}}}{6MF}$

$$16\pi^2\,\mu\frac{dC_{221014}}{d\mu}=\beta_{C_{221014}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column		Representative term structure
T1		$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T2		$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3		$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$	$\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$